

Phase I/II Statistical Process Monitoring SECOM Data

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統計專論 (III) Final Report

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Project Objective

- Dataset: UCI SECOM
- Objective: Construct SPC workflow

Phase I → Control chart design → Phase II

- Main comparison: Shewhart, EWMA, CUSUM, and CPD charts
- Final:
 1. Which chart signals first ?
 2. Whether the pattern suggests mean shift, variance shift, or structural change ?

Selected Variable and Preprocessing

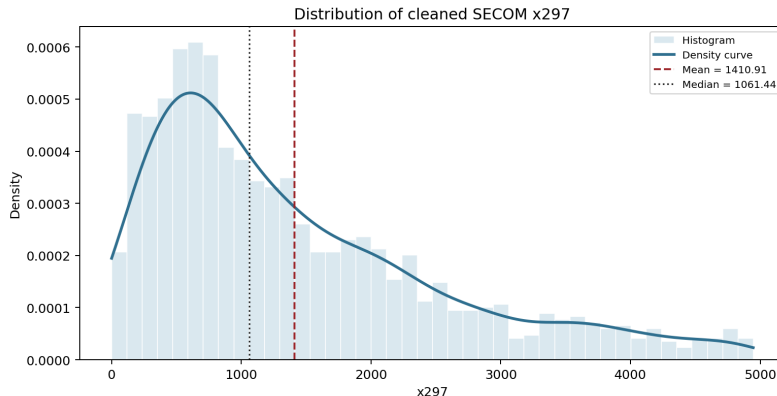
Variable

Selected process variable: x_{297}

- Reason: Low missing, High distinct
- Samples: 1567
- Missing samples removed: 2
- Outlier: 1.5 IQR filter after dropping missing values
- Outliers removed: 128
- Cleaned samples: **1437**

Variable Summary

Mean	SD	Min	Q1	Median	Q3	Max
1410.91	1126.51	0	556.99	1061.44	1995.39	4945.92



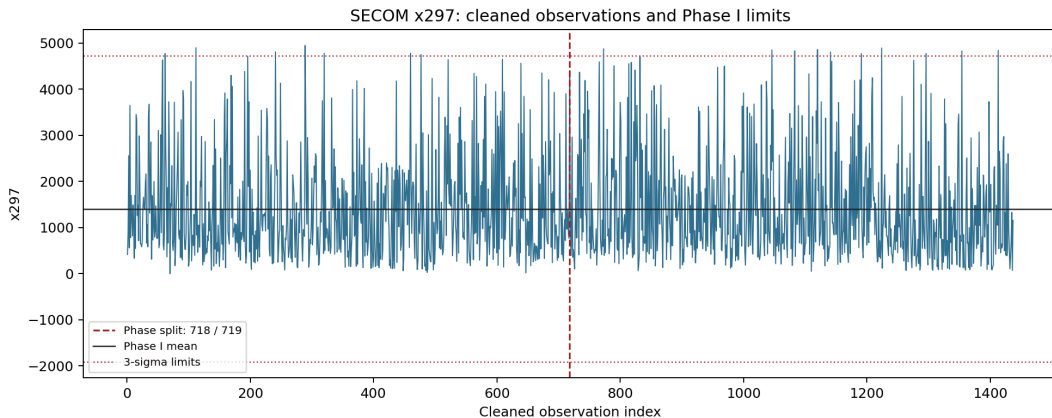
Phase I / Phase II Split

$$m = \lfloor 0.5n \rfloor = \lfloor 0.5(1437) \rfloor = 718$$

Phase	samples	Mean / SD
Phase I	718	1403.64 / 1105.53
Phase II	719	1418.18 / 1147.80

- Phase I is used as the in-control reference sample
- Phase II is monitored sequentially using the chart limits estimated from Phase I

Cleaned Series and Phase Split



Phase I RS/P Diagnosis

Recursive segmentation + permutation result

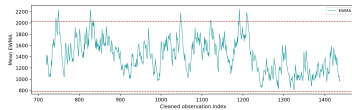
$$p_{\text{level}} = 0.8675, \quad p_{\text{scale}} = 0.8325$$

- Phase I shows no significant evidence of level or scale change
- Phase I data can be regarded as stable

Method	Target	Main setting
Individuals Shewhart	Mean	3-sigma limits
Mean EWMA	Mean	$\lambda = 0.1, \rho = 2.454$
Variance EWMA	Variance	$\lambda = 0.1, \rho_U = 2.595, \rho_L = 1.580$
Adaptive EWMA	Mean	eta1 case (ii), $h = 0.7874$
Mean CUSUM	Mean	$k = 0.5, h = 4.095$
Variance CUSUM	Variance	$k_U = 1.848, h_U = 7.416; k_L = 0.462, h_L = -2.446$
CPD charts	Mean / variance / joint	$\alpha = 0.005$

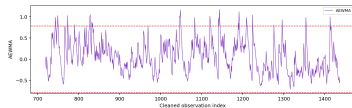
- EWMA and CUSUM settings follow the in-class R code.
- CPD limits follow Chapter 6 approximations for mean, variance, and joint charts.

Mean-Shift Monitoring Results (Main Mean Charts)



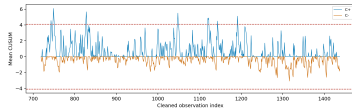
First signal: 743
Signal count: 17

Mean EWMA



First signal: 743
Signal count: 32

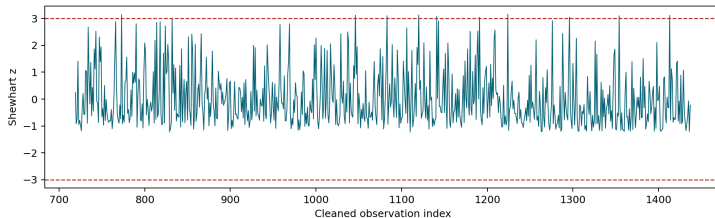
Adaptive EWMA



First signal: 743
Signal count: 13

Mean CUSUM

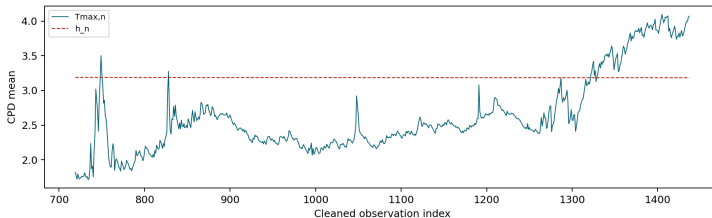
Mean-Shift Monitoring Results (Supplementary Charts)



Individuals Shewhart

First signal: 773

Signal count: 10

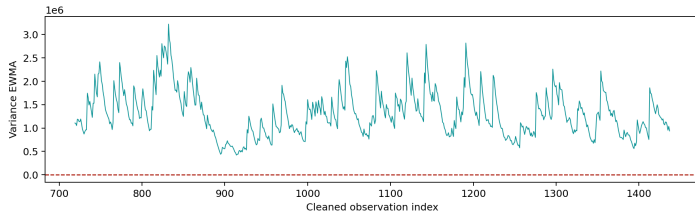


CPD mean chart

First signal: 749

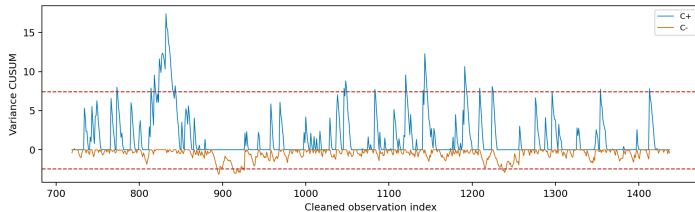
Signal count: 118

Variance-Shift Monitoring Results (Target: Variance)



Variance EWMA

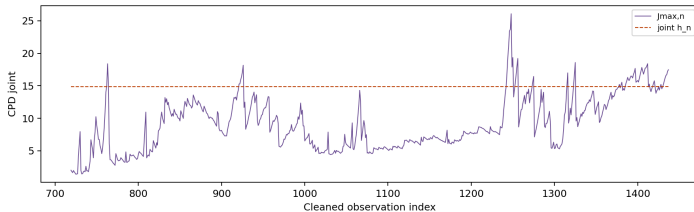
First signal: 719
Signal count: 719



Variance CUSUM

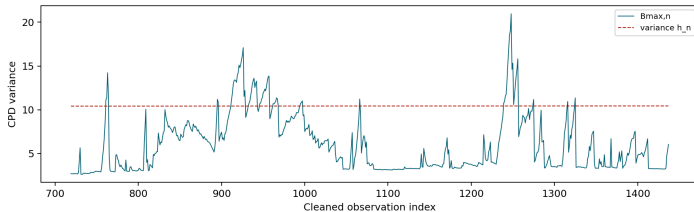
First signal: 773
Signal count: 59

CPD and Joint Monitoring Results



CPD joint chart

First signal: 762
Signal count: 71



CPD variance chart

First signal: 761
Signal count: 78

CPD Post-Signal Interpretation

- CPD mean chart first signals at index 749
- CPD variance chart first signals at index 761
- CPD joint chart first signals at index 762
- The strongest CPD variance and joint signals estimate the likely change location around:

$$\hat{r} \approx 1229.$$

- The strongest CPD mean signal estimates:

$$\hat{r} \approx 1212$$

- Phase I RS/P diagnosis does not reject stability, so the reference sample is usable.
- Phase II mean changes are detected by EWMA, AEWMA, CUSUM, and CPD mean chart.
- Variance-sensitive methods are more aggressive, especially variance EWMA.
- CPD variance and joint charts locate later structural changes around indices 1229.
- Overall pattern suggests not a single isolated mean shift only, but a combination of mean instability and variance/structural change.

Conclusion

- ① The selected SECOM variable x_{297} is usable after missing-value removal and IQR outlier filtering.
- ② Phase I RS/P results support using the first 718 cleaned samples as the IC reference.
- ③ The earliest signal is variance EWMA at the first Phase II observation.
- ④ For mean-shift detection, mean EWMA, AEWMA, and mean CUSUM signal earliest at index 743.
- ⑤ CPD charts provide additional localization and suggest later structural changes around indices 1212–1229.